

ORIE 3120: Mapping the Payoff: How Major, College, and Region Shape Career Salaries

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Introduction:

What factors determine how much a college graduate earns 10 years into their career? While common wisdom often focuses on major or starting salary, other aspects, like where you attend college and which region you live in, may also play a significant role. This project investigates four interrelated questions about long-term salary growth using data from PayScale Inc., as reported by the Wall Street Journal. We explore how region, field of study, starting vs. growth-focused majors, and college type influence mid-career earnings.

Our questions are designed to provide practical value to students, families, and education advisors. Should a student prioritize a higher starting salary or a field with long-term growth? Do Liberal Arts graduates really catch up to STEM majors? Is it worth attending a more prestigious college? These are non-obvious, high-stakes questions with real financial consequences.

To answer them, we use statistical techniques including bootstrapping, confidence intervals, forecasting, and linear regression. All analyses are grounded in real-world survey data and interpreted with clarity for non-technical audiences.

Dataset Description

We use three datasets from the Wall Street Journal, originally compiled by PayScale Inc., based on self-reported survey data from over 1.2 million U.S. college graduates. These datasets focus exclusively on individuals with only a bachelor's degree, making them especially relevant to undergraduate career planning. Each dataset captures starting salaries, mid-career salaries (around 10 years post-grad), and percent change between these points.

- **Degrees That Pay Back:** Indexed by undergraduate major, it contains fields like Starting Median Salary, Mid-Career Median Salary, % Change, and Salary Percentiles (10th, 25th, 75th, 90th).
- **Salaries by College Type:** Indexed by school name and type (e.g., Ivy League, Party, Engineering), with the same salary metrics.
- **Salaries by Region:** Indexed by school and geographic region, used to compare regional wage trends.

To support our analysis, we cleaned all salary fields, converting dollar strings to floats, and in some questions we joined datasets using shared fields like school name or major. We also categorized majors into broader groups (e.g., STEM, Humanities, Business) for better generalization in some analyses.

Because this data is self-reported, there may be some sampling bias, particularly if higher earners are more likely to participate. It also does not account for graduate degrees or cost of living. However, it remains one of the most comprehensive publicly available snapshots of salary outcomes by major and institution.

This dataset provides the foundation for our four questions, each of which explores a different, but connected, dimension of career earnings. Together, they offer a nuanced understanding of what shapes financial outcomes in the decade after college graduation.

Research Questions:

- How do starting and mid-career salaries vary by university region, and which areas offer the best salary growth?
- Do different fields (humanities, STEM, etc) show significantly different patterns of salary growth?
- Is it better to choose a major with a high starting salary or one with higher long-term salary growth?
- Does college type (engineering vs party vs liberal arts vs ivy) matter more than major for mid-career salary?

Our research is especially relevant for students choosing majors, families considering the cost of different colleges, career advisors, and policymakers focused on education and workforce planning. Understanding how different educational and geographic factors impact long-term salaries can help individuals make more informed decisions about their education and career paths.

Question 1: *How do starting and mid-career salaries vary by university region, and which areas offer the best salary growth?*

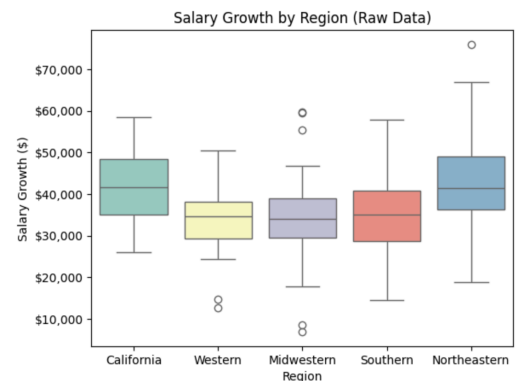
This analysis investigates how starting and mid career salaries vary by university region, with a focus on which areas offer the best salary growth. Our null hypothesis is that all regions have the same mean salary growth. Our alternative hypothesis is that at least one group has a different mean salary growth. To test whether salary growth varies significantly by region, we conducted a p-value hypothesis test. To find which regions differ and by how much, we used bootstrap to create a confidence interval.

Analysis of Raw Data

We calculated the salary growth by calculating “Median Starting Salary” - “Median Mid-Career Salary”.

Figure 1:

The box plot illustrates the salary growth by region with the cleaned data. Generally, there is similar salary growth across all regions. However, based on the graph, the northeastern region has a slightly higher salary growth, and the midwestern region has a slightly lower salary growth. To test whether the salary growth by region is statistically significant, we conducted a p-value hypothesis test of each region’s salary.



Hypothesis test of Salary Growth By Region

We found that the p-value, the probability of observing the differences in salary growth across regions if the null hypothesis were true, is 2.1082e-12. This value is much smaller than 0.05 which means we reject the null hypothesis. See **Appendix** for explanation on 0.05 threshold. Therefore, each region's salary growth is significantly different.

Data Analysis: Bootstrapping

The p-value hypothesis test analysis told us that there is a statistically significant difference in salary growth across regions. However it doesn't tell us how large the differences are, how uncertain the estimates are, and which specific regions differ the most. To resolve these gaps, we conduct bootstrap analysis to get the confidence interval, and check assumptions since bootstrap is non parametric.

Figure 2:

For each region, we simulate 300 alternative samples from that region’s data with replacement. We calculate the mean salary growth of each sample which gives us a distribution of possible means. From there we compute the bootstrap mean and confidence interval. As we can see in the bar plot of the bootstrapped means salary growth by region, California and Northeastern have significantly greater salary growth compared to other regions.

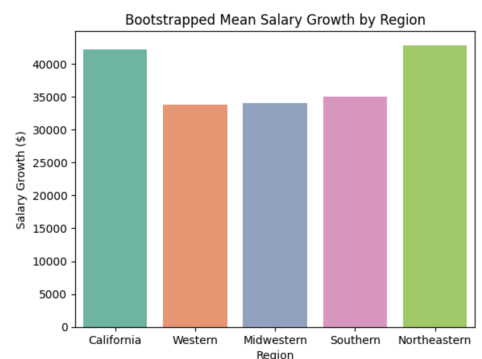


Figure 3:

Our bootstrapped confidence interval table shows that Northeastern has the highest estimated mean growth (\$42,835), followed closely by California (\$42,248). Midwestern and Western regions have the lowest mean growth (around \$33,900–\$34,000), and the Southern region is in the middle at ~\$34,964. The Confidence intervals of California and the Northeastern regions do not overlap with the Midwestern or Western regions, suggesting those differences are likely statistically significant.

Bootstrapped Confidence Intervals:

	Region	Mean Growth	Lower CI (2.5%)	Upper CI (97.5%)
0	California	42200.916667	39383.839286	44934.017857
1	Midwestern	34024.849765	32072.042254	35954.014085
2	Northeastern	42845.936667	41177.900000	44723.750000
3	Southern	35055.113924	33284.177215	37251.202532
4	Western	33844.150794	31701.428571	35913.154762

Conclusion:

Our results show that we reject our null hypothesis and conclude that salary growth does differ by region. We know this is statistically significant because our p value is 2.1082e-12 which is much lower than 0.05. To understand which regions' salary growth differed and by how much who conducted bootstrap to create a confidence interval. We found that Northeastern has the highest estimated mean growth (\$42,835), followed closely by California (\$42,248). Therefore, one can make an educated prediction on their salary growth based on the region of the university they attend.

Question 2: Do different fields (humanities, STEM, etc) show significantly different patterns of salary growth?

Building on our geographic analysis, we next asked whether a student's chosen academic field also significantly affects salary growth. This analysis investigates whether a student's academic field such as STEM, Business, Humanities, or Arts, has an effect on salary growth. While STEM degrees are often assumed to lead the best long term financial outcomes, this question is meant to test this assumption by comparing not just average growth but the consistency of outcomes within each field. Our null hypothesis is that the mean percent salary growth is the same across all major categories. Our alternative hypothesis is that at least one major category has a different mean percent salary growth. This analysis is useful for students choosing a major and wondering whether certain fields offer better or more guaranteed returns on their educational investment.

Figure 1:

This first figure we created shows a view of how the salary ranges vary within their respective fields. This figure shows how salary growth varies within each academic field. Business and Humanities show high growth; however, they have wider ranges and whiskers implying that the results are more spread out. Some students in these majors do very well but others may see less growth. STEM majors tend to see more consistent results as they have a smaller range. Lastly people in Arts Majors tend to have a wide range showing more variation. Overall this figure shows that while other Major groups may have higher growth and rewards, STEM is a more stable choice.

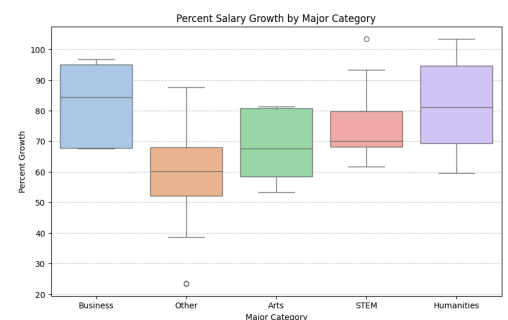
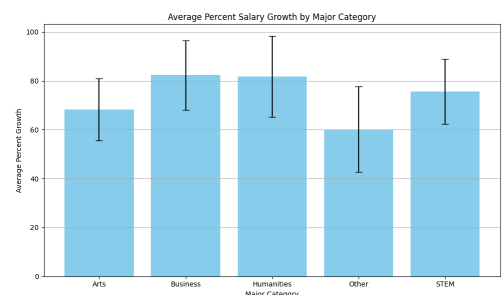


Figure 2:

The bar chart shows the average salary growth in each field. Arts and Humanities have the highest average growth but also the largest spread, while STEM is a more stable choice.



meaning outcomes can vary widely. Business and STEM fields show slightly lower average growth but are more stable, with smaller error bars. This reflects more predictable returns over time. Overall this chart helps show how some majors may offer higher growth later on but also come with a risk of ranges.

Data Analysis: Linear Regression

To better understand whether a major category significantly impacts salary growth, we performed a linear regression model.

The model produced an R-squared value of 0.272, meaning that approximately 27.2% of the variation in salary growth can be explained by a student's academic field. While this is not high, it confirms that the field of study has an impact.

While the regression shows some differences between fields, none of the coefficients are statistically significant at the 0.05 level, so we can't say with high confidence that the field alone causes

higher or lower growth. However, none of the individual categories are statistically significant at the 0.05 level meaning that while we may find an effect by the groups overall collectively, no single field itself differs strongly to reject the hypothesis of no effect with high confidence. Still, the general trend suggests that Arts and Humanities may have higher upside potential, while STEM fields appear more reliable but with slightly lower long-term growth.

OLS Regression Results						
Dep. Variable:	Percent change from Starting to Mid-Career Salary				R-squared:	0.272
Model:	OLS				Adj. R-squared:	0.268
Method:	Least Squares				F-statistic:	4.210
Date:	Tue, 13 May 2025				Prob (F-statistic):	0.00558
Time:	03:20:37				Log-Likelihood:	-286.76
No. Observations:	50				AIC:	423.5
DF Residuals:	45				BIC:	433.1
DF Model:	4					
Covariance Type:	nonrobust					
	coef	std err	t	P> t	[0.025	0.975]
const	68.2400	7.138	9.571	0.000	53.880	82.600
Business	14.0600	10.083	1.394	0.170	-6.248	34.368
Humanities	13.4886	9.335	1.445	0.155	-5.313	32.290
Other	-8.1357	7.867	-1.034	0.307	-23.980	7.709
STEM	7.4100	8.732	0.849	0.401	-10.177	24.997
Omnibus:	0.105				Durbin-Watson:	1.952
Prob(Omnibus):	0.949				Jarque-Bera (JB):	0.158
Skew:	-0.098				Prob(JB):	0.924
Kurtosis:	2.887				Cond. No.	8.50

Conclusion

Our results show that the academic field does impact salary growth. Business and Humanities majors tend to experience higher growth, while STEM majors see more stable and predictable returns. Arts and Other fields have less consistent outcomes, making them potentially riskier choices. Overall, we reject the null hypothesis. There is statistical evidence that salary growth differs by major category, meaning the field of study does have an effect on how much salaries grow from early to mid-career. When choosing a major, students should consider both expected salary growth and the consistency of outcomes. A major with high average returns but high variability may not suit everyone. Meanwhile, majors like Business or STEM offer more reliable growth, which could better support predictable financial planning.

Question 3: Is it better to choose a major with a high starting salary or one with higher long-term salary growth?

While major explains some variation, we wanted to understand whether starting salary or growth over time better predicts long-term returns. To get a more definitive answer to whether majors with high starting salary or high long-term salary growth percentage is a more important factor in long-term outcomes, we used a linear regression model to predict a variable we defined as the "return" for each major, discussed in more detail below. This approach allows us to dive deeper into the analysis between these two variables and quantify the impacts of each predictor.

In existing literature, there is evidence that suggests the relationship between income and quality-of-life outcomes (such as longevity) is "concave". In other words, logarithmic models better capture the relationship between life quality and income due to the diminishing return effect. Specifically, in study [1], it was found that life expectancy could be modeled

as a logarithmic function of income. Using these findings as a foundation, we defined a target variable "return" metric that would be roughly proportional to this life expectancy/quality of life metric. We defined $\text{Return} = \log(15 * \text{Starting Median Salary} + 35 * \text{Mid-Career Median Salary})$. It weighs the mid-career salary more heavily and assumes the average American works for ~50 years before retirement [2].

For our linear model, we used two predictors: (1) Starting (2) Percent Change from Starting to Mid-Career Salary Percent Growth) to estimate our constructed "return" value. non-linear relationships in these predictors and address patterns that arose in our residuals, we applied the transformations to the predictors: For Starting Median logarithmic transformation, followed by a square root to diminishing returns; for Percent Growth, we introduced polynomial to the 6th power to flexibility capture the relationship.

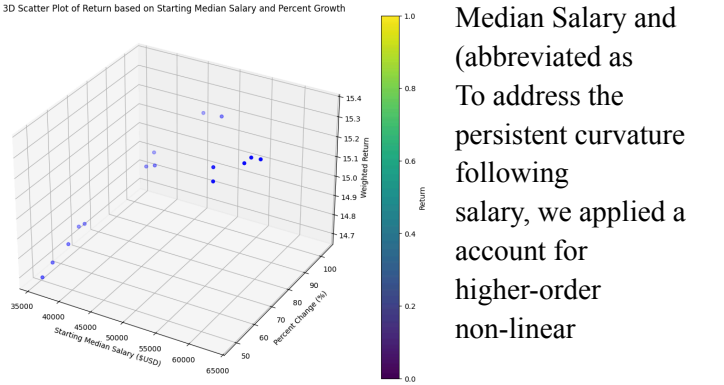


Figure 1: 3D Scatter Plot of Return based on Starting Median Salary and Mid-Career Growth Percentage

It is important to note the limitations of modelling with a linear regression. First, it assumes that these two variables are fully independent, which may not be fully true in the real world since some high-paying majors might experience more complex growth trends that are beyond the scope of our project.

To avoid overfitting and assess model performance, we used a train-test split where we reserved 30% of the data for testing, and we evaluated accuracy using the root mean squared error (RMSE). When testing the quality of our model we found that both the RMSE for the train and test were both 0.0525. The closeness of the values suggests the model generalizes well to unseen data. Additionally, the small value compared to the scale of the return variable also indicates strong predictive accuracy.

OLS Regression Results						
Dep. Variable:	Returns	R-squared:	0.985			
Model:	OLS	Adj. R-squared:	0.984			
Method:	Least Squares	F-statistic:	1069.			
Date:	Tue, 06 May 2025	Prob (F-statistic):	5.00e-30			
Time:	03:07:03	Log-Likelihood:	84.217			
No. Observations:	35	AIC:	-162.4			
Df Residuals:	32	BIC:	-157.8			
Df Model:	2					
Covariance Type:	nonrobust					
	coef	std err	t	P> t	[0.025	0.975]
const	13.6940	0.028	480.863	0.000	13.636	13.752
Starting Median Salary	2.078e-05	4.65e-07	44.698	0.000	1.98e-05	2.17e-05
Percent Change	0.0053	0.000	24.359	0.000	0.005	0.006
Omnibus:	19.681	Durbin-Watson:	2.488			
Prob(Omnibus):	0.000	Jarque-Bera (JB):	30.835			
Skew:	-1.427	Prob(JB):	2.01e-07			
Kurtosis:	6.605	Cond. No.	3.26e+05			

Conclusion

Our final model suggests that percentage growth from beginning to mid-career salary has a larger marginal impact on long-term “return” or our proxy for quality of life. Specifically, we end up finding that the median salary has a coefficient of 0.02078 and growth has a coefficient of 0.053, implying that growth has a larger impact per unit increase in return. This provides insight for students weighing the decision of a major that would best maximize their long-term outcomes in terms of life quality. However, when we simulated optimal values based on our prediction model, we found that the optimal starting median salary was \$63,200, and the optimal percentage growth was 69.3% to maximize the “return” variable. These values are interesting since the median salary and percentage growth in the data based on U.S. college graduates were ~\$40,000 and ~68%, respectively. This suggests that most degrees already deliver growth near the ideal range, so in contrast, the true real-world starting salary is lower than what is the “optimal” threshold. With all these factors in mind, students should prioritize majors that offer higher-than-average starting salaries, without compromising growth potential.

Question 4: Does college type (engineering vs party vs liberal arts vs ivy) matter more than major for mid-career salary?

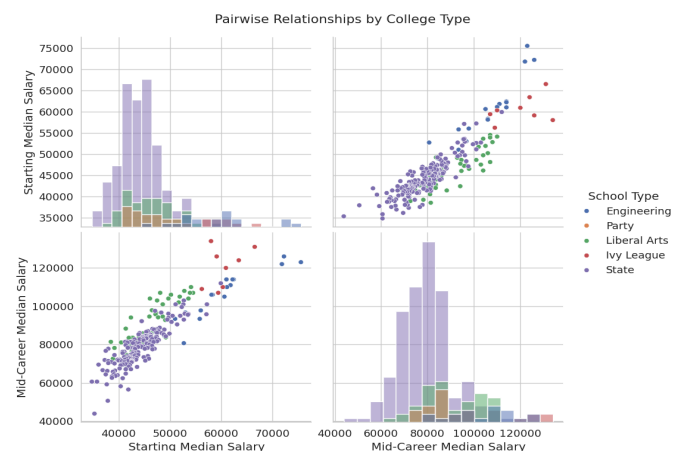
To complete the picture, we examined whether institutional factors like school type outweigh major factors in influencing salary growth. This analysis investigates whether the type of college attended (Ivy League, Liberal Arts, Party School, Engineering, or State School) is a better predictor of mid-career salary than a student's major. Our guiding hypothesis is that institutional context might shape long-term salary growth even more than field of study. We explore this through regression analysis and a series of visualizations using data from the Wall Street Journal (via PayScale).

Exploratory Visualization: Pairwise Relationships

We begin with a pairplot showing starting and mid-career salaries across different school type

Figure 1: Pairwise Salary Distributions by School Type

- Starting salaries are higher for Engineering and Ivy League schools.
- Mid-career salaries show more divergence, especially favoring Ivy League and Liberal Arts schools.
- State and Party Schools have a wider spread, suggesting less predictable salary growth.



These patterns hint that **school type** could drive mid-career outcomes beyond the effect of starting salary or major alone. This motivates a closer look via regression.

Data Analysis: Linear Regression

The OLS regression output reinforces our earlier observations from the exploratory analysis. The model achieves a high R-squared of 0.852 and adjusted R-squared of 0.849, suggesting that the combination of starting salary and college type explains a large portion of the variability in mid-career salaries across graduates.

Key findings from the regression coefficients:

- **Starting Median Salary** is statistically significant with a p-value less than 0.001 and a coefficient of 2.02. This means that for every additional dollar in starting salary, we expect about \$2 more in mid-career salary — a strong and predictable relationship. The very small p-value tells us this result is extremely unlikely to have occurred by random chance, and we can confidently reject the null hypothesis that starting salary has no effect.

OLS Regression Results						
Dep. Variable:	Mid-Career Median Salary	R-squared:	0.852			
Model:	OLS	Adj. R-squared:	0.849			
Method:	Least Squares	F-statistic:	301.8			
Date:	Tue, 29 Apr 2025	Prob (F-statistic):	1.01e-106			
Time:	02:23:47	Log-Likelihood:	-2699.1			
No. Observations:	269	AIC:	5410.			
Df Residuals:	263	BIC:	5432.			
Df Model:	5					
Covariance Type:	nonrobust					
	coef	std err	t	P> t	[0.025	0.975]
const	-1.07e+04	3347.796	-3.195	0.002	-1.73e+04	-4103.279
Starting Median Salary	2.0229	0.075	26.877	0.000	1.875	2.171
Is_Ivy	8485.9286	2361.575	3.593	0.000	3835.929	1.31e+04
Is_Liberal_Arts	7533.1562	924.026	8.153	0.000	5713.727	9352.586
Is_Party	2903.7779	1321.379	2.198	0.029	301.949	5505.607
Is_Engineering	-4930.3188	1754.004	-2.811	0.005	-8383.996	-1476.642
Omnibus:	6.626	Durbin-Watson:	1.317			
Prob(Omnibus):	0.036	Jarque-Bera (JB):	8.695			
Skew:	-0.176	Prob(JB):	0.0129			
Kurtosis:	3.888	Cond. No.	5.22e+05			

- **Ivy League** graduates earn, on average, \$8,486 more in mid-career salary than similar students from state schools, controlling for starting salary and major. The p-value is less than 0.001, meaning this difference is statistically significant — we can be confident it's not due to noise in the data. In plain terms, this supports the idea that attending an Ivy League school leads to real, measurable long-term financial advantages.
- **Liberal Arts** college graduates experience a similarly strong premium of \$7,533 ($p < 0.001$), again showing that institutional type, not just major, plays a powerful role in salary growth.
- **Party School** graduates receive a \$2,903 boost in mid-career salary ($p = 0.029$). Since this p-value is below 0.05, the result is still considered statistically significant, though we interpret it with slightly more caution than the larger effects above.
- **Engineering School** graduates, surprisingly, see a negative effect of $-\$4,930$ compared to state schools ($p = 0.005$). While these graduates often start with higher salaries, the lower coefficient here suggests they experience slower salary growth over time. The p-value confirms that this finding is statistically significant, and we can again reject the null hypothesis and interpret the trend as meaningful.

In summary, each of these p-values tells us that the differences we observe between school types are highly unlikely to be due to random chance. Instead, they point to a consistent and statistically sound pattern: school type has a major independent effect on long-term salary outcomes, even after accounting for a student's starting salary and major. This evidence supports our hypothesis that institutional prestige and educational context can shape earnings potential well beyond graduation.

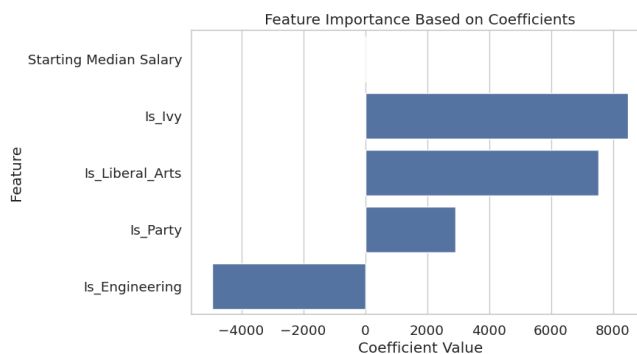
Assumption checks for the linear regression model, including residual vs. fitted values and normality of residuals, were conducted and confirmed that the model meets key OLS assumptions. See **Appendix** Figures 2 and 3 for full diagnostics.

Feature Importance: Coefficient Visualization

Figure 4: Feature Importance Based on Coefficients

The bar plot of regression coefficients reveals:

- **Ivy League** and **Liberal Arts** variables have the largest positive impact on mid-career salary growth.
- **Engineering** school dummies have a surprising negative contribution when controlling for starting salary.
- **Starting salary** is still influential but relatively less compared to school type effects.



This clearly shows that institutional prestige (college type) matters significantly for mid-career outcomes.

Conclusion

Our results reveal that college type significantly influences mid-career salary growth, even after controlling for starting salary and major. Attending an Ivy League or Liberal Arts college leads to substantially higher mid-career earnings, likely due to networking, reputation, or employer perception. Engineering schools, while advantageous early in a career, may not

offer the same long-term payoff. This challenges assumptions about the primacy of major in career success. Where you go to college, and the reputation associated with it, can matter even more than what you study.

Final Conclusion

Our analysis uncovered meaningful insights about the long-term impact of educational decisions. Through four interrelated and compelling questions, we explored how factors such as region, academic discipline, salary growth vs. starting salary, and school type influence mid-career earnings. These questions are not only statistically interesting, but personally and socially relevant, they influence how students and families navigate one of the most important investments in their lives: college.

In Question 1, we found that regional differences persist, with California and the Northeast leading in salary growth, likely due to economic density and industry clustering. Question 2 revealed that broad academic fields vary in growth potential, with Humanities surprisingly leading in percentage growth, even if their starting salaries are lower. Question 3 demonstrated that choosing a major for long-term growth yields greater payoffs than chasing high starting salaries alone. And in Question 4, we discovered that college type, particularly Ivy League and Liberal Arts schools, has a larger impact on mid-career salary growth than major, challenging conventional narratives about the supremacy of STEM.

Throughout the project, we applied a diverse set of analysis techniques, from confidence interval estimation and bootstrapping, to regression modeling and variable comparison. We were careful to check assumptions, validate results with visualizations, and communicate findings in a way that connects with non-technical audiences. Each method was chosen not just for technical correctness, but because it helped answer a real-world question in a compelling way.

Ultimately, this project shows that no single variable – major, region, or school – wholly determines salary outcomes. Instead, it's the interaction of these factors that shapes career earnings. For students and their families, this means thinking holistically about tradeoffs. For educators and policymakers, it calls for more nuanced, data-informed advising frameworks.

These insights are not just academic, they can directly inform how young people plan their futures. And that, in our view, is what makes this project valuable.

Appendix

M1 Notebook: <https://colab.research.google.com/drive/1cyhyfEaIY1WApW-0TQ5omKQl0QuoNdKq?usp=sharing>

M2 Notebook: <https://colab.research.google.com/drive/1JTmf2DeC7F3qzQblFKJyINALY7fQJOmM?usp=sharing>

Question 1 p-value Hypothesis Test

The p value is a number, calculated from a statistical test, that describes how likely you are to find a specific set of observations if the null hypothesis is true. p values are used in hypothesis testing to help decide whether to reject the null hypothesis. The smaller the p value, the more likely you are to reject the null hypothesis. Generally when the p value is less than 0.05, it is small enough to be considered statistically significant and reject the null hypothesis.

Question 2 Assumption Checks:

Figure 1: Each major category included a sufficient number of data points and the distributions were reasonable supporting the use of the boxplot

Figure 2: The means of each group were sufficiently representative with roughly symmetric group distributions. This justifies the use of error bars.

Linear Regression: The linear regression model showed no significant issues with linearity, variance, or normality so the OLS assumptions were satisfied.

Question 3: Assumption Checks

Checking Linear in Parameters:

After initially checking residual plots using a LOWESS line for when I plotted the variables linearly, there was curvature in both residuals, leading me to do the transformations discussed in question 3. There are still some random shapes, but that likely comes from the limitations of the linear model. But not significant curvature or pattern in either.

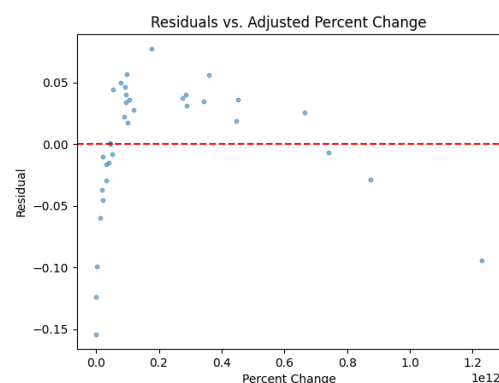
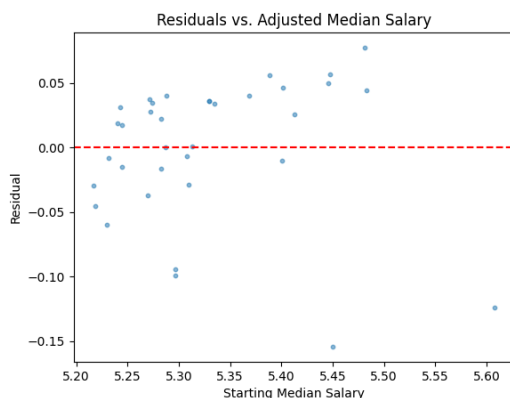


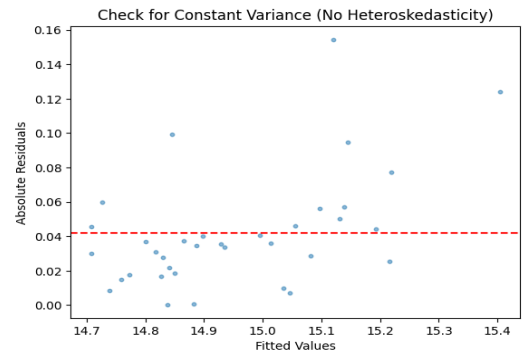
Figure 1a: Adjusted Starting Median Salary Residuals

Figure 1b: Adjusted Percent Change Residuals

Checking Constant Variance (No Heteroskedasticity):

When plotting the absolute residuals versus fitted values, the points appear generally randomly scattered, with a minor increase in scattering for large values. However, with the small number of data points and lack of a strong fanning outwards shape, or upward trend, it generally suggests that the assumption of constant variance has been reasonably satisfied. It's important to acknowledge that the logical and structural limitations of linear regression make capturing relationships at extremely large values in our dataset challenging. The 4-5 points that create the somewhat “spanning” out effects are likely due to a more complex relationship that was beyond the scope of our project.

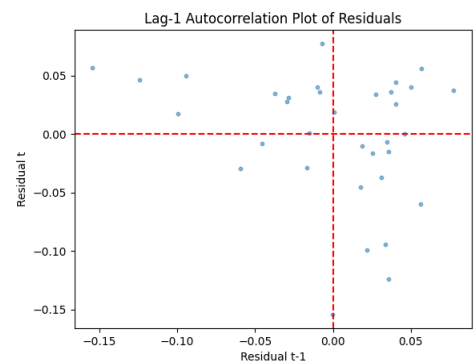
Figure 2: Absolute Residuals vs Fitted Values



Checking Independence:

Figure 3: Lag-1 Autocorrelation Plot of Model Residuals

The residuals are clustered randomly around both axes, indicating that there is no systematic structure or clustering in the errors. This supports our assumption described in Question 3 section that the prediction errors for Median Starting Salary and Growth Percentages are not dependent on each other. While in the modeling the two predictors as independent might still not be entirely accurate, the lack of autocorrelation suggests that our transformations to each predictor (log and polynomial) sufficiently captured the joint effects.



Checking Normality:

The residuals are roughly normal, but a slight departure in the tails, however as mentioned before the real-world limitations and potential weak relationships between the variables is likely the cause. The center, where the model is most sensitive, shows that it is reasonably well-satisfied.

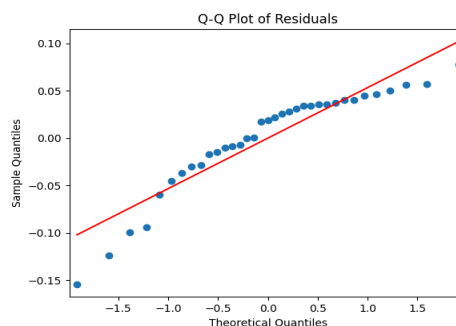
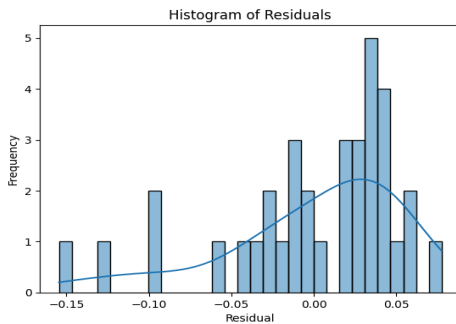


Figure 4a: Histogram of Model Residuals

Figure 4b: Q-Q Plot of Residuals

Studies Cited in Question 3:

- [1] <https://pmc.ncbi.nlm.nih.gov/articles/PMC4866586/>
- [2] <https://www.deputy.com/blog/how-americans-spend-their-time>

Question 4 Assumption Checks: Residual Analysis

Figure 2: Residual vs Fitted Values shows no major curvature or fanning, supporting linearity and constant variance assumptions.

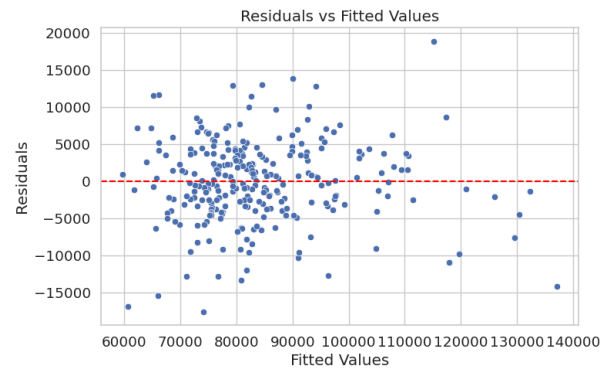
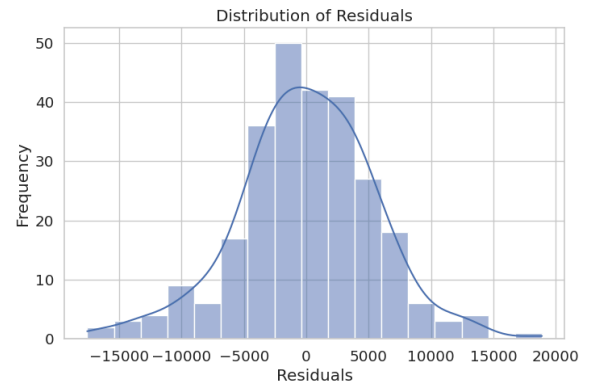


Figure 3: Distribution of Residuals approximates a bell curve, suggesting residuals are normally distributed.



These diagnostic plots validate the appropriateness of the OLS model.